DOMESTIC AIRLINES — TRAINING

Approach Operations and Minima

Course handbook

Classification of approach operations (2D/3D, Type A/B, CAT I–III), aerodrome operating minima, ceiling and visibility, the approach ban and LVO.

4 lessons · 1 modules · about 0.5 hours

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.

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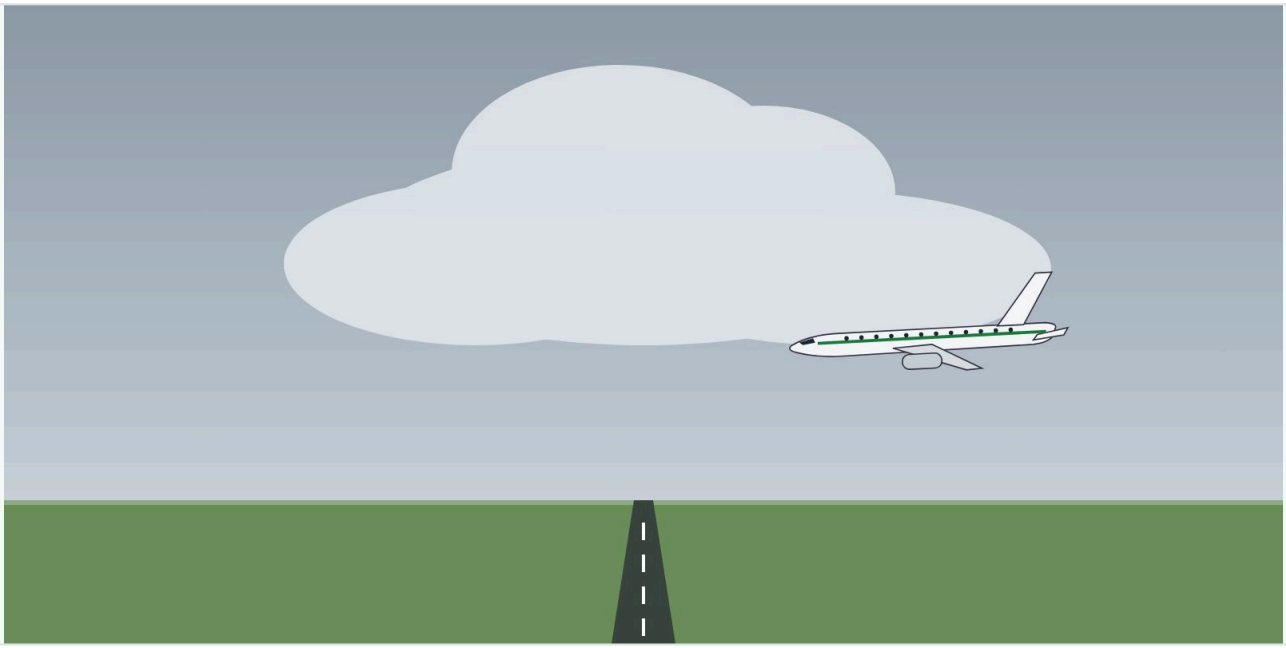
MODULE 1 · APPROACH OPERATIONS

Lesson 01: Classification of Instrument Approach Operations

Lesson 1 of 4 · about 7 min

LEARNING OBJECTIVES

- Classify instrument approach operations as 2D or 3D, Type A or Type B.
- State the CAT I, CAT II and CAT III criteria.
- Identify which operations need a specific approval.



Type A or Type B — know what you are flying.

2D and 3D

- **2D** operation — lateral guidance only (e.g. VOR, NDB, LOC, LNAV, LP), flown to an **MDA/H**, using the CDFA technique.
- **3D** operation — lateral and vertical guidance (e.g. ILS, GLS, LNAV/VNAV, LPV), flown to a **DA/H**.

Type A and Type B (ICAO Annex 6, adopted by EASA)

Type	Minimum height	Category	Criteria
Type A	MDH or DH at or above 250 ft	—	Non-precision and some 3D approaches with higher minima

Type B	DH below 250 ft	CAT I	DH not lower than 200 ft, and visibility not less than 800 m or RVR not less than 550 m
		CAT II	DH lower than 200 ft but not lower than 100 ft, and RVR not less than 300 m
		CAT III	DH lower than 100 ft or no DH, and RVR less than 300 m or no RVR limitation

The former subdivisions CAT IIIA, IIIB and IIIC are no longer used: all are now **CAT III**. Operations below CAT I (CAT II, CAT III), and some operational credits, need a **specific approval** (EASA SPA.LVO).

COMPANY AUTHORISATION

The approach operations each fleet is authorised for — for example CAT I only, or CAT II for a fleet and crew with the approval — are listed in the **AOC operations specifications** issued by the **DACM**. The method and values for minima are in **OM A**; the aerodrome minima are in **OM C** and the charts.

KEY TAKEAWAYS

- 2D to MDA/H (CDFA); 3D to DA/H.
- Type A: 250 ft or above. Type B: below 250 ft — CAT I, II, III.
- CAT I: DH ≥ 200 ft, VIS ≥ 800 m or RVR ≥ 550 m.
- CAT II / III need a specific approval.

KNOWLEDGE CHECK

An ILS with a DH of 200 ft and RVR 550 m is classified as:

A. Type A

B. Type B, CAT I

C. Type B, CAT II

D. CAT III

Answer: B. DH 200 ft and RVR 550 m are the CAT I limits, which is a Type B operation.

MODULE 1 · APPROACH OPERATIONS

Lesson 02: Aerodrome Operating Minima

Lesson 2 of 4 · about 7 min

LEARNING OBJECTIVES

- Explain how aerodrome operating minima are established.
- Distinguish DA/H, MDA/H, RVR, visibility and ceiling.
- Apply take-off and circling minima.



Minima: DA/H or MDA/H, RVR or visibility.

Establishing minima

The operator establishes aerodrome operating minima (EASA CAT.OP.MPA.110) for each aerodrome, taking into account: the aircraft type and performance, the crew, the runway characteristics and lighting, the navigation and visual aids, obstacles (the OCA/H of the procedure), the type of approach and the means of reporting the weather. The minima **must not be lower** than those established by the State of the aerodrome, unless specifically approved.

The elements

- **DA/H** — the altitude or height at which a go-around must be started on a 3D approach if the required visual reference has not been established.
- **MDA/H** — the minimum on a 2D approach (with CDFA, a go-around is started at or slightly above it as the company specifies).
- **RVR** — runway visual range, measured along the runway; **VIS** — reported meteorological visibility.
- **CMV** — converted meteorological visibility: a reported visibility converted to an RVR-equivalent value when RVR is not available, using the company table.

- **Ceiling** — used mainly for **planning** (destination and alternate) and for circling and some non-precision approaches as the OM A specifies.

Take-off and circling

- **Take-off minima** are expressed as RVR or visibility, sometimes with a ceiling — and depend on the runway lighting and the aircraft's ability to return or reach a take-off alternate.
- **Circling minima** (MDH and visibility) depend on the aircraft approach category (A to D). Keep visual reference with the runway throughout.

COMPANY AUTHORISATION

The approach operations each fleet is authorised for — for example CAT I only, or CAT II for a fleet and crew with the approval — are listed in the **AOC operations specifications** issued by the **DACM**. The method and values for minima are in **OM A**; the aerodrome minima are in **OM C** and the charts.

KEY TAKEAWAYS

- Operator minima: never lower than the State's, unless approved.
- DA/H for 3D, MDA/H for 2D; RVR, VIS, CMV; ceiling mainly for planning.
- Take-off and circling minima are separate — check them.

KNOWLEDGE CHECK

May the operator's minima be lower than those published by the State of the aerodrome?

A. Yes, always

B. No, unless specifically approved

C. Only at night

D. Only for CAT I

Answer: B. Minima must not be lower than the State's, unless specifically approved.

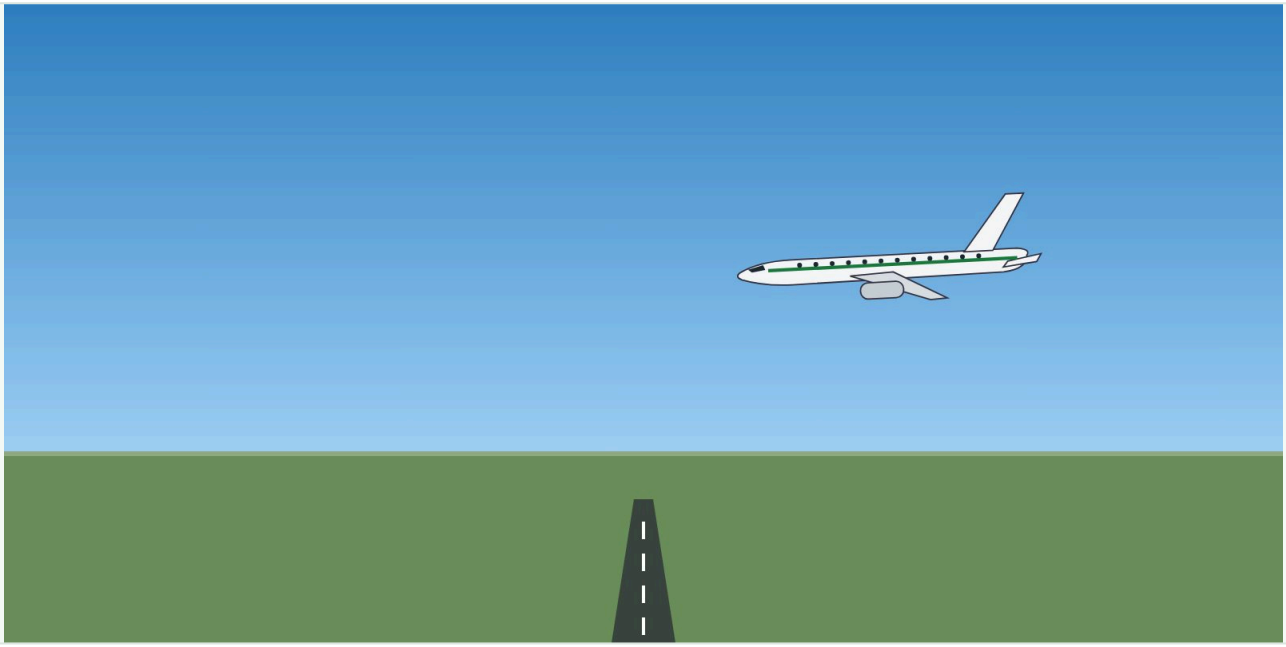
MODULE 1 · APPROACH OPERATIONS

Lesson 03: Planning Minima and the Approach Ban

Lesson 3 of 4 · about 6 min

LEARNING OBJECTIVES

- Apply planning minima for destination and alternates.
- Apply the approach ban rule.
- Make the go-around decision at the minimum.



Approach ban: 1,000 ft above the aerodrome.

Planning

At the planning stage, the weather at the destination and the alternates must be at or above the **planning minima** for the expected time of use (often with a time window before and after the ETA). Planning minima for alternates are higher than the landing minima — the values are in OM A (based on CAT.OP.MPA.185 / 245).

The approach ban

APPROACH BAN (EASA CAT.OP.MPA.305)

The commander shall not continue an approach **below 1,000 ft above the aerodrome** (or into the final approach segment) if the reported RVR or visibility is **less than the applicable minimum**. If it falls below the minimum after passing 1,000 ft, the approach may be continued to the DA/H or MDA/H.

At the minimum

- Continue below the DA/H or MDA/H only if the **required visual reference** is established and can be maintained — for example elements of the approach lights, the threshold, the touchdown zone or their lights.
- Otherwise, go around — and fly the published missed approach.
- If visual reference is lost below the minimum, go around immediately.

COMPANY AUTHORISATION

The approach operations each fleet is authorised for — for example CAT I only, or CAT II for a fleet and crew with the approval — are listed in the **AOC operations specifications** issued by the **DACM**. The method and values for minima are in **OM A**; the aerodrome minima are in **OM C** and the charts.

KEY TAKEAWAYS

- Plan with planning minima (higher than landing minima) for destination and alternates.
- Approach ban: no continuing below 1,000 ft if reported RVR / VIS is below minima.
- Below DA/MDA only with the required visual reference.

KNOWLEDGE CHECK

At 1,500 ft above the aerodrome, the reported RVR drops below your minimum. What applies?

A. Continue to the DA anyway

B. The approach ban: do not continue below 1,000 ft above the aerodrome

C. Land without reference

D. Request CAT III

Answer: B. Below 1,000 ft the approach may not be continued if the reported RVR is below the minimum.

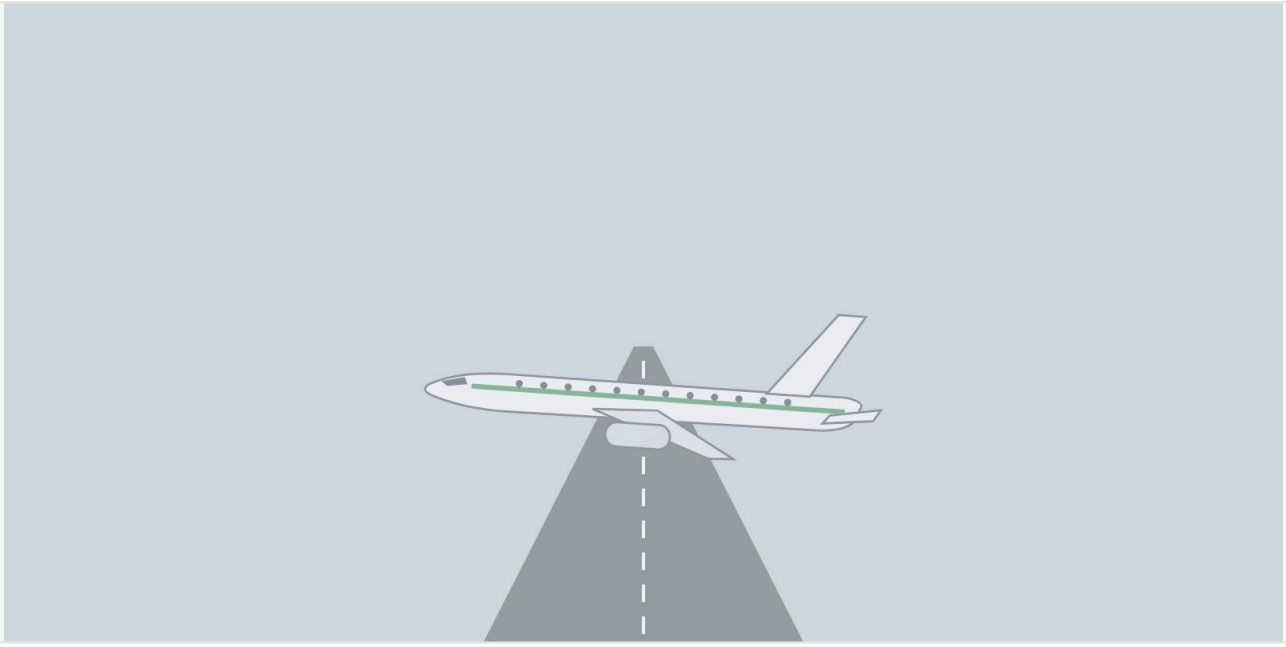
MODULE 1 · APPROACH OPERATIONS

Lesson 04: Low-Visibility Operations Overview

Lesson 4 of 4 · about 6 min

LEARNING OBJECTIVES

- Describe the requirements for CAT II and CAT III operations.
- Explain low-visibility procedures at aerodromes.
- Recognise when LVO capability is lost.



CAT II and III need a specific approval.

Requirements

- A **specific approval** (SPA.LVO) for the operator, with the aircraft certified, the crew trained and qualified, and procedures in the Operations Manual.
- The aerodrome with **low-visibility procedures (LVP)** in force, and the required ground equipment serviceable.
- The aircraft equipment required for the category serviceable — check the MEL and the AFM.

Low-visibility procedures

When LVP are in force, ATC protects the ILS sensitive and critical areas, increases spacing and uses specific taxi routes. Low-visibility take-off (LVTO) below the standard take-off minima also needs approval and suitable runway lighting.

Loss of capability

- Aircraft failures (autopilot channels, autothrottle, displays) can downgrade the approach category — the FCOM / AOM gives the effect.

- Ground equipment failures can also raise the minima — see the next course on inoperative ground equipment.
- Fog: know the types (radiation, advection) and their trends for planning.

COMPANY AUTHORISATION

The approach operations each fleet is authorised for — for example CAT I only, or CAT II for a fleet and crew with the approval — are listed in the **AOC operations specifications** issued by the **DACM**. The method and values for minima are in **OM A**; the aerodrome minima are in **OM C** and the charts.

KEY TAKEAWAYS

- CAT II / III: approval, aircraft, crew, procedures, aerodrome LVP and equipment.
- LVP protect ILS areas and increase spacing.
- Aircraft or ground failures can downgrade the category.

KNOWLEDGE CHECK

Which is NOT required to fly a CAT II approach?

A. Operator specific approval

B. LVP in force at the aerodrome

C. Trained and qualified crew

D. A weather radar image of the fog

Answer: D. A weather radar image is not a requirement.

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